

Flight Operations Dashboard (Delay Analysis Focus)

2015 U.S Domestic Flights

Which page do you want to view now?



Overview



Root Cause Analysis



Airline & Airport
Performance



Time Pattern
Analysis

Key insights:

In 2015, the U.S. airline system operated **5.82 million flights**, of which **1.02 million flights were delayed (17.9%)** and **89,884 flights were cancelled (1.54%)**. The **on-time rate reached 81.9%**, indicating a relatively stable system overall, while still exposing **significant operational risk**.

Total delay amounted to **62.65 million minutes**, with an **average arrival delay of approximately 60.6 minutes** for delayed flights (> 15 minutes), demonstrating that delays were **severe in nature**, rather than minor schedule deviations.

72.0% of total delay minutes originated from controllable factors, primarily:

- **Late Aircraft Delay: 39.8%**
- **Airline-related Delay: 32.2%**

In contrast, **Weather, National Airspace System (NAS), and Security delays collectively accounted for only 27.9%** of total delay minutes.

Delay patterns also exhibit strong **temporal concentration**:

- **32.1% of delayed flights occurred during peak hours (16:00–19:59)**.
- **Summer (30.9%) and Winter (27.4%) together accounted for more than 58% of total annual delay minutes**.

Delays were **not evenly distributed across the system**:

- Large network carriers such as **Southwest Airlines, American Airlines, and Delta Air Lines** contributed the highest volume of delays due to their **scale and hub concentration**.
- Several airlines with **lower reliability and higher delay severity**, including **Spirit Airlines, Frontier Airlines, and American Eagle Airlines**, should be **prioritized for operational review**.
- **Ground congestion is heavily concentrated at major hub airports**, notably **John F. Kennedy International Airport, LaGuardia Airport, Newark Liberty International Airport, and Chicago O'Hare International Airport**, where **average taxi-out times for delayed flights exceed 30 minutes**.



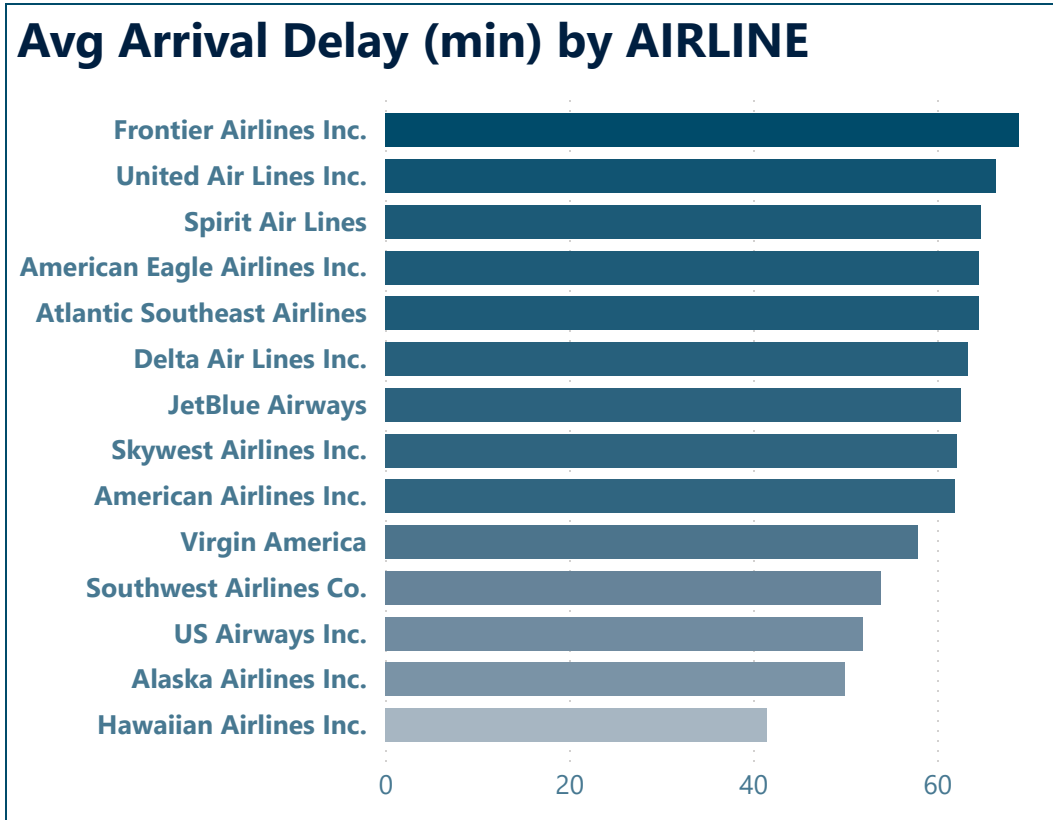
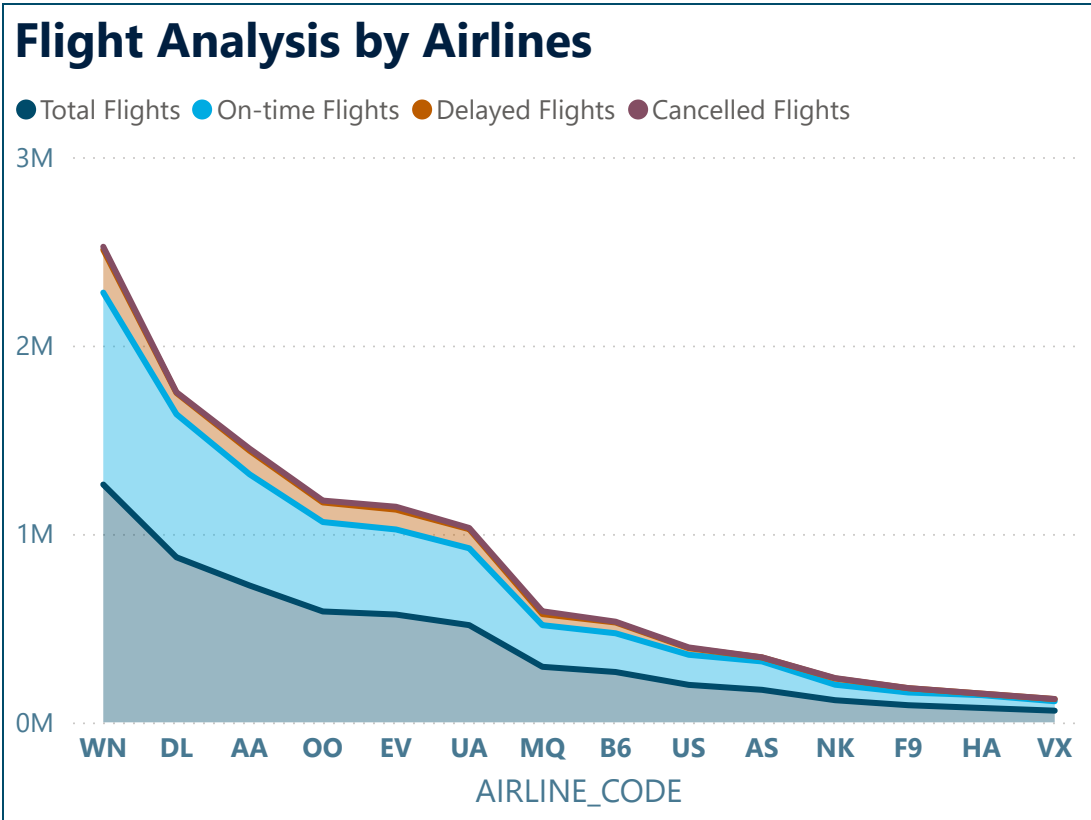
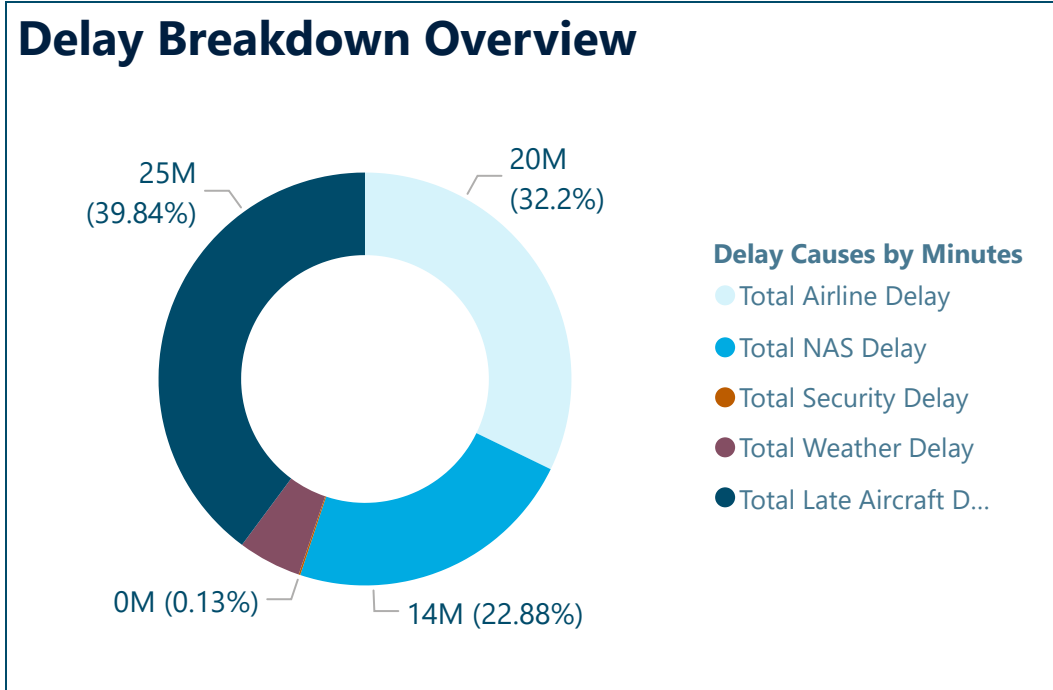
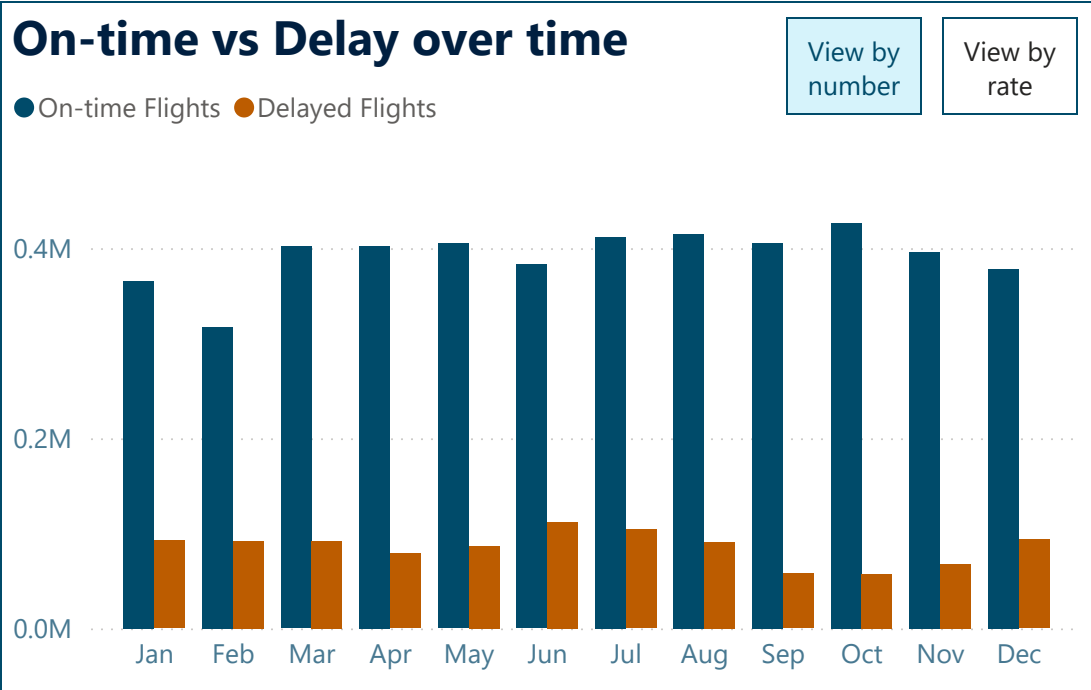
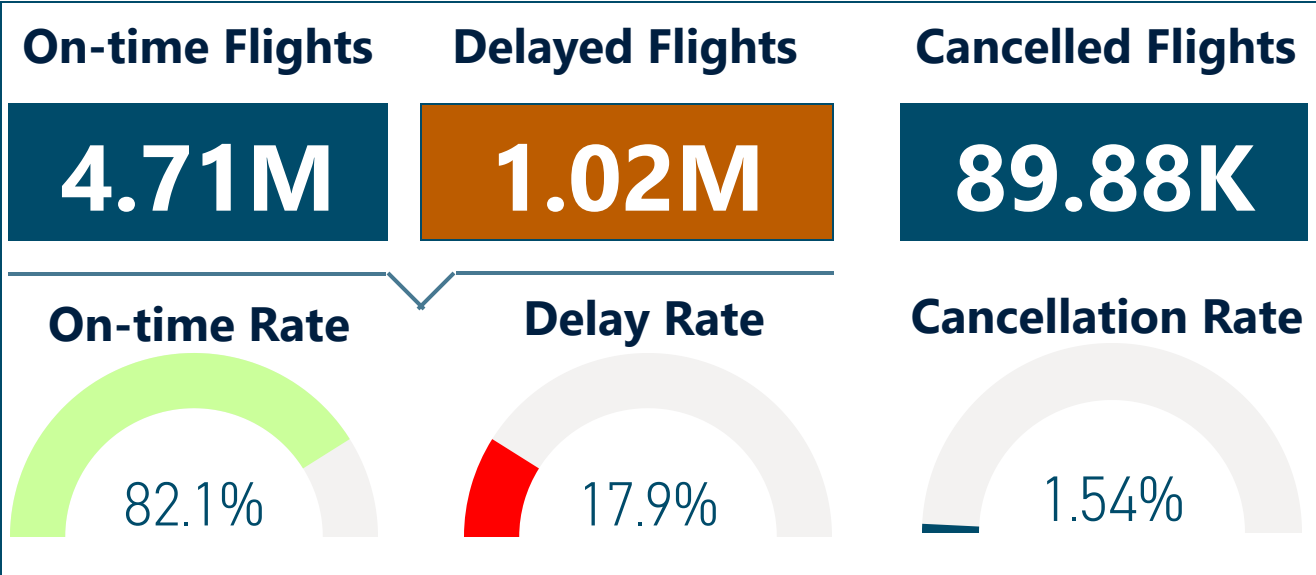
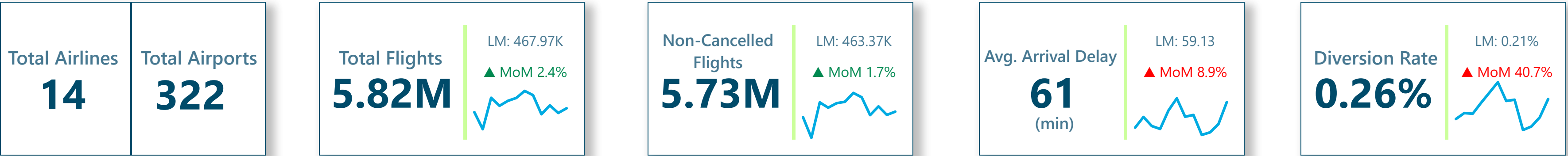


2015 Airline Delay Analysis - Overview

MONTH
All

AIRLINE
All

IATA_CODE
All



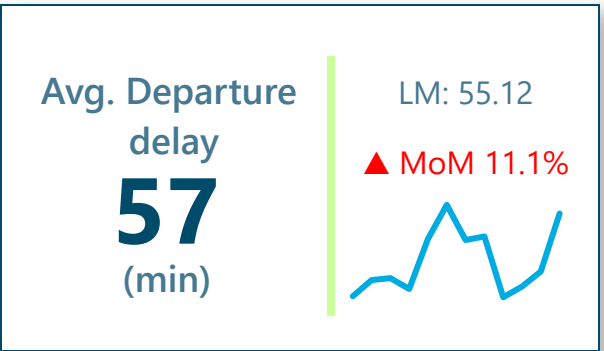
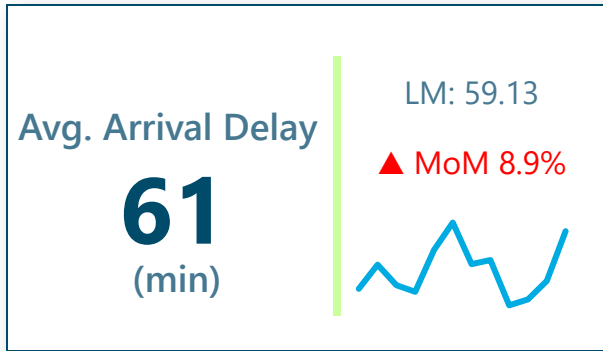
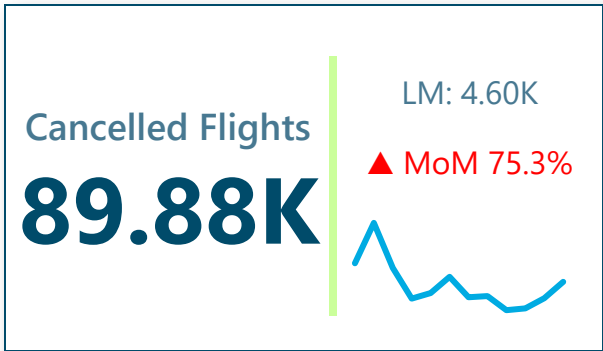
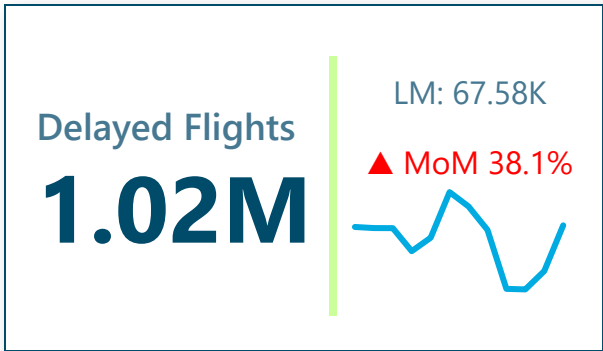
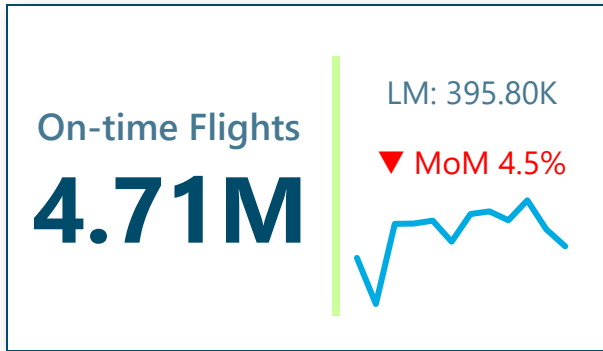


2015 Airline Delay Analysis - Delay Root Causes

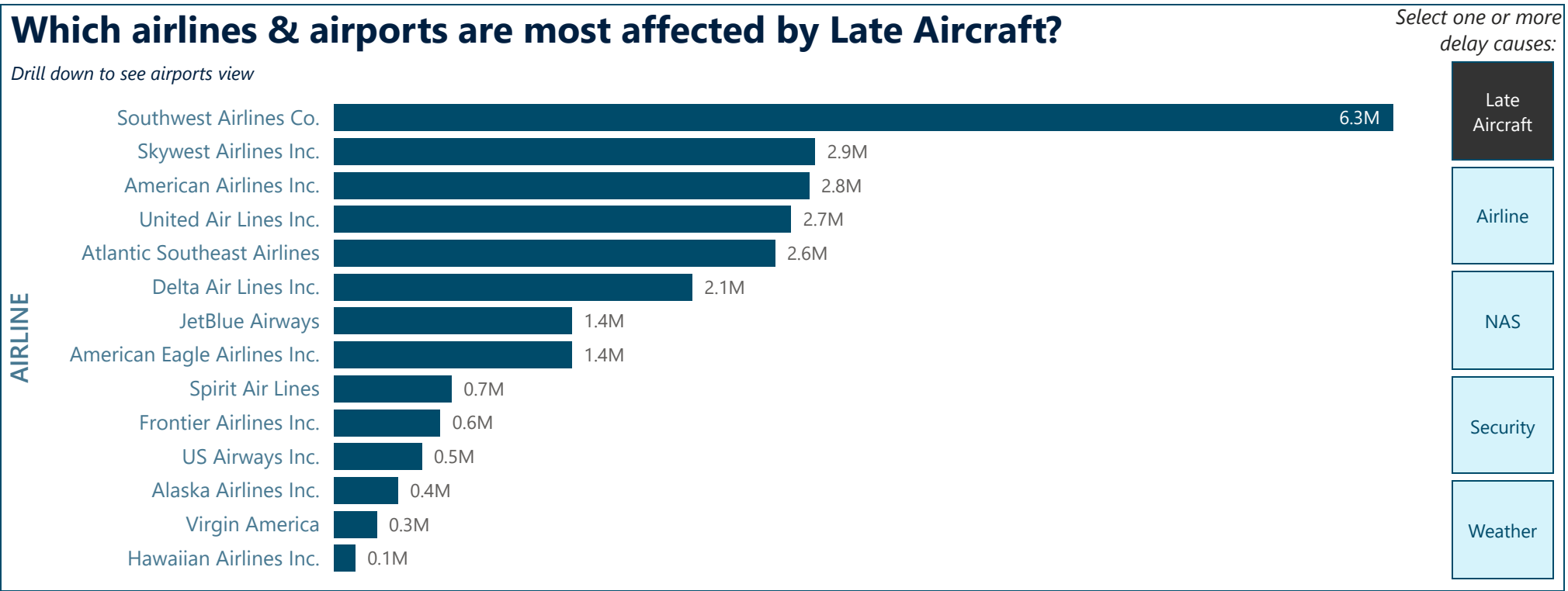
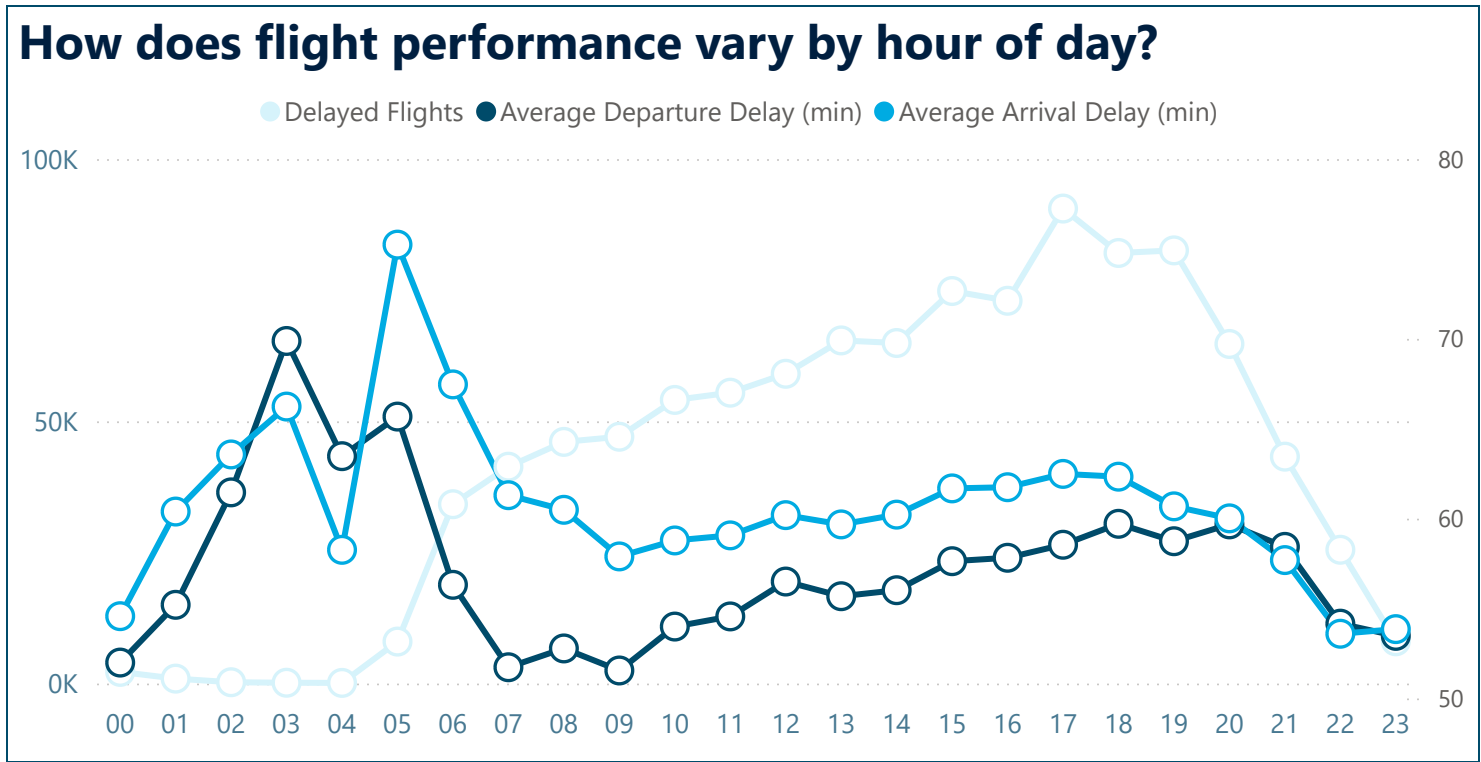
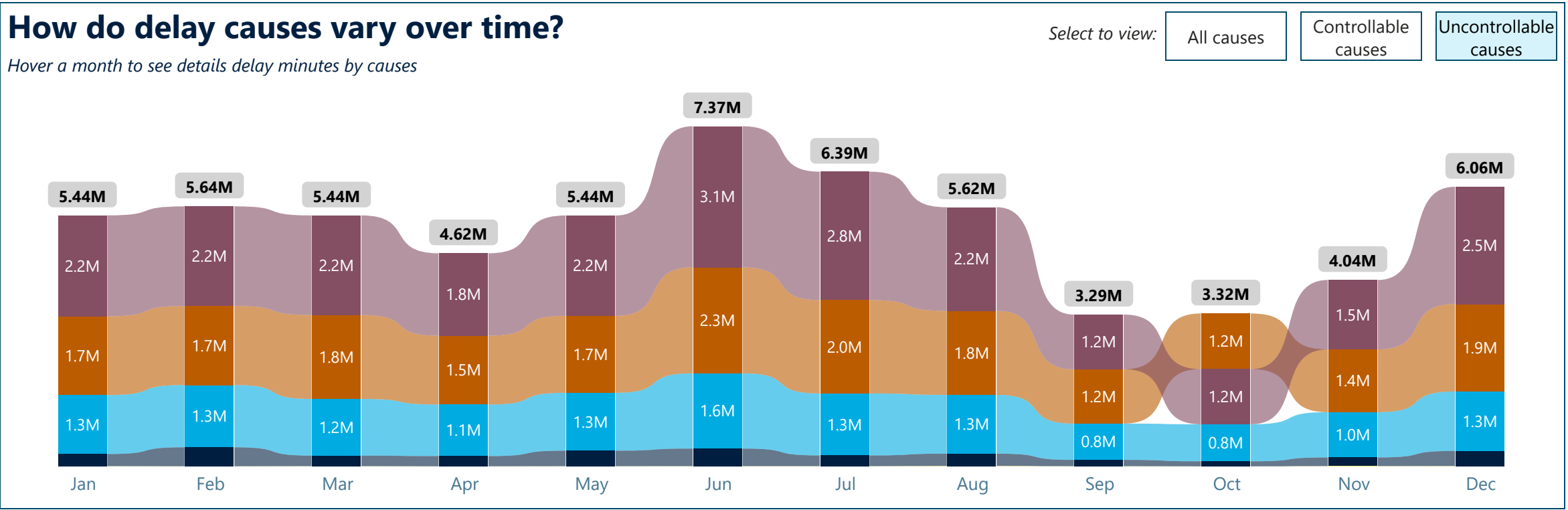
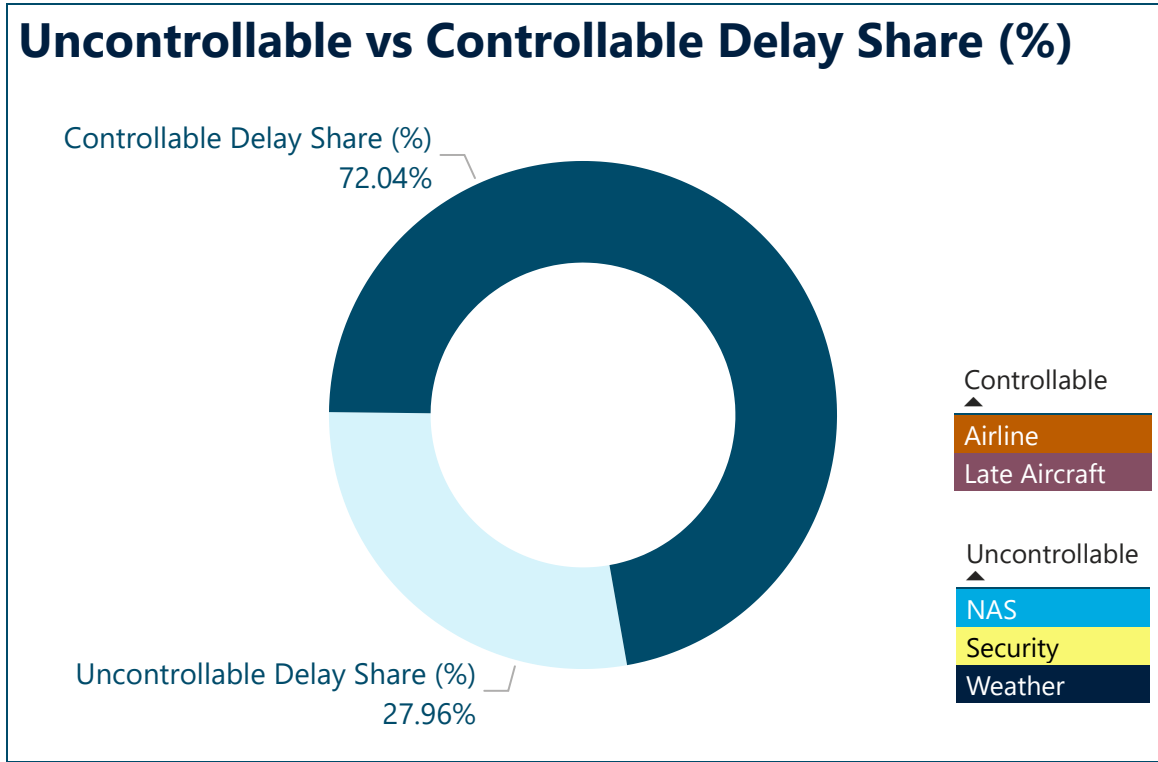
MONTH
All

AIRLINE
All

IATA_CODE
All



⚠ Attention: Delay causes are measured in total delay minutes to reflect severity rather than frequency (number of flights).





2015 Airline Delay Analysis - Airline and Airport Operations

MONTH

▼

All

▼

AIRLINE

▼

All

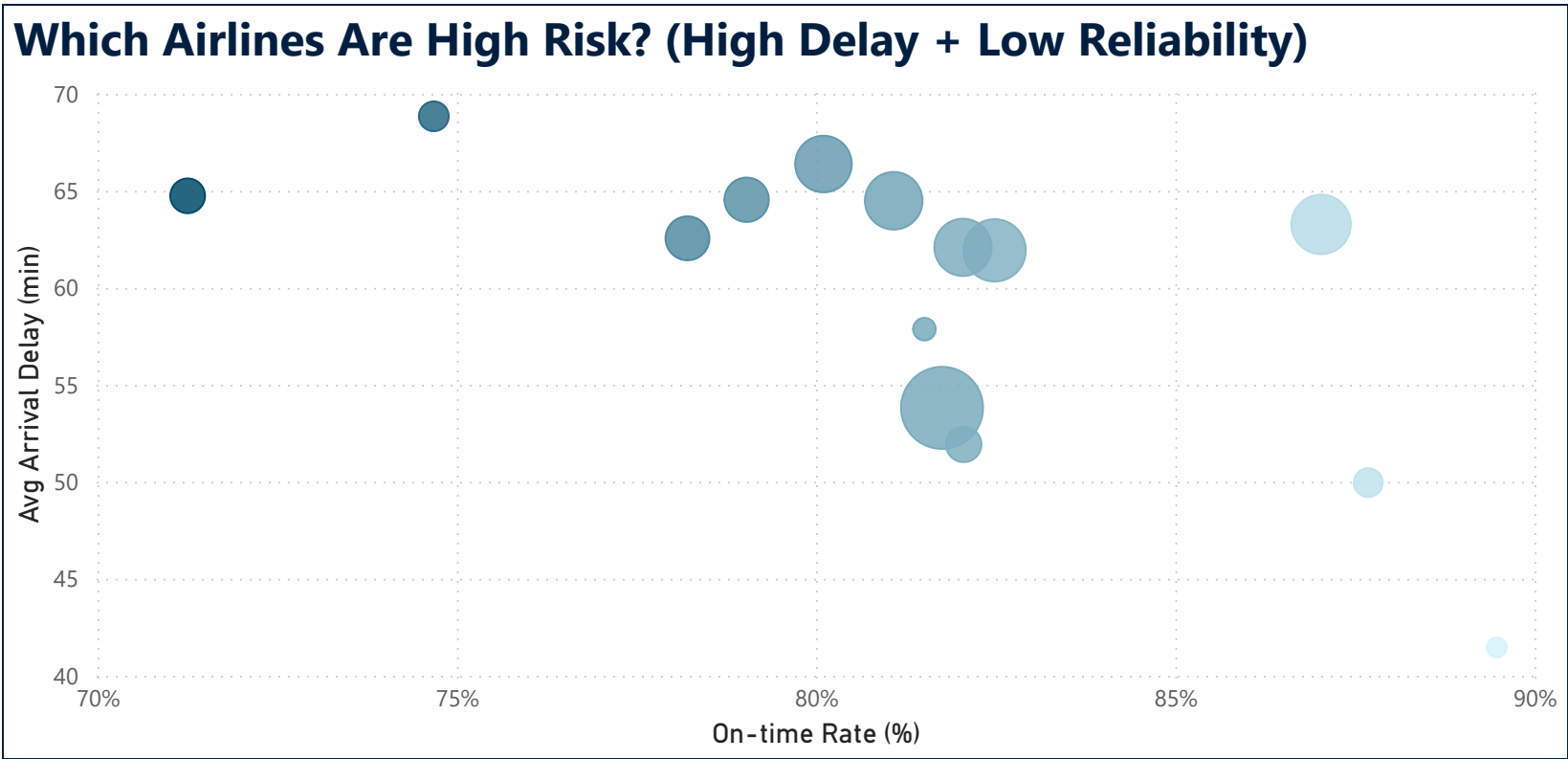
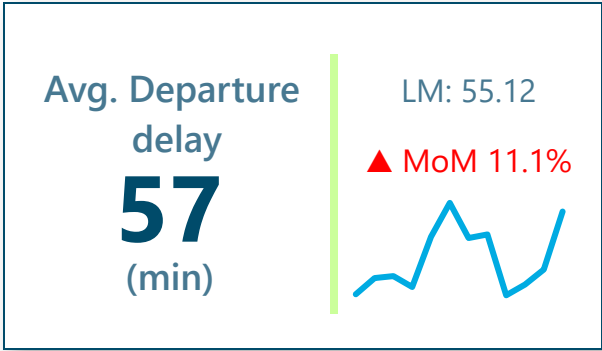
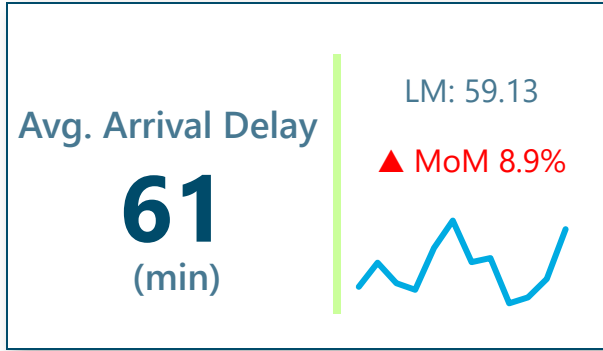
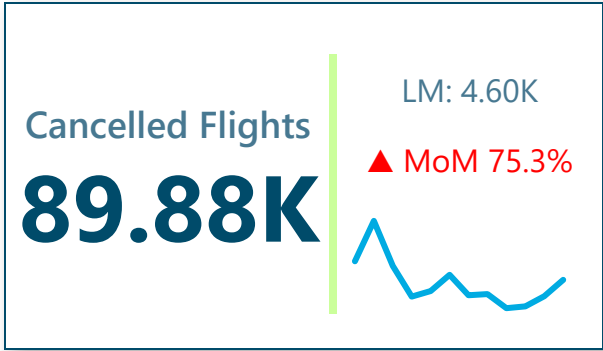
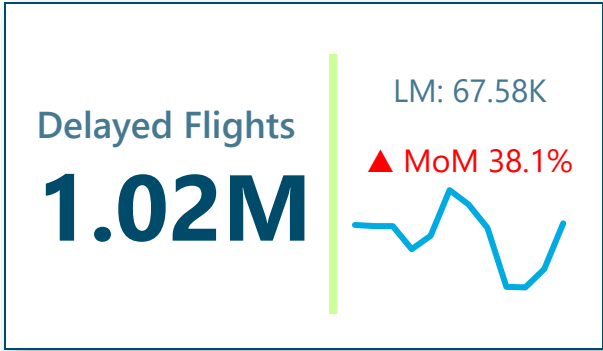
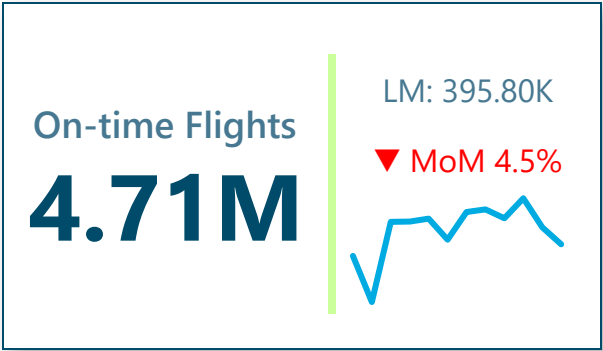
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All

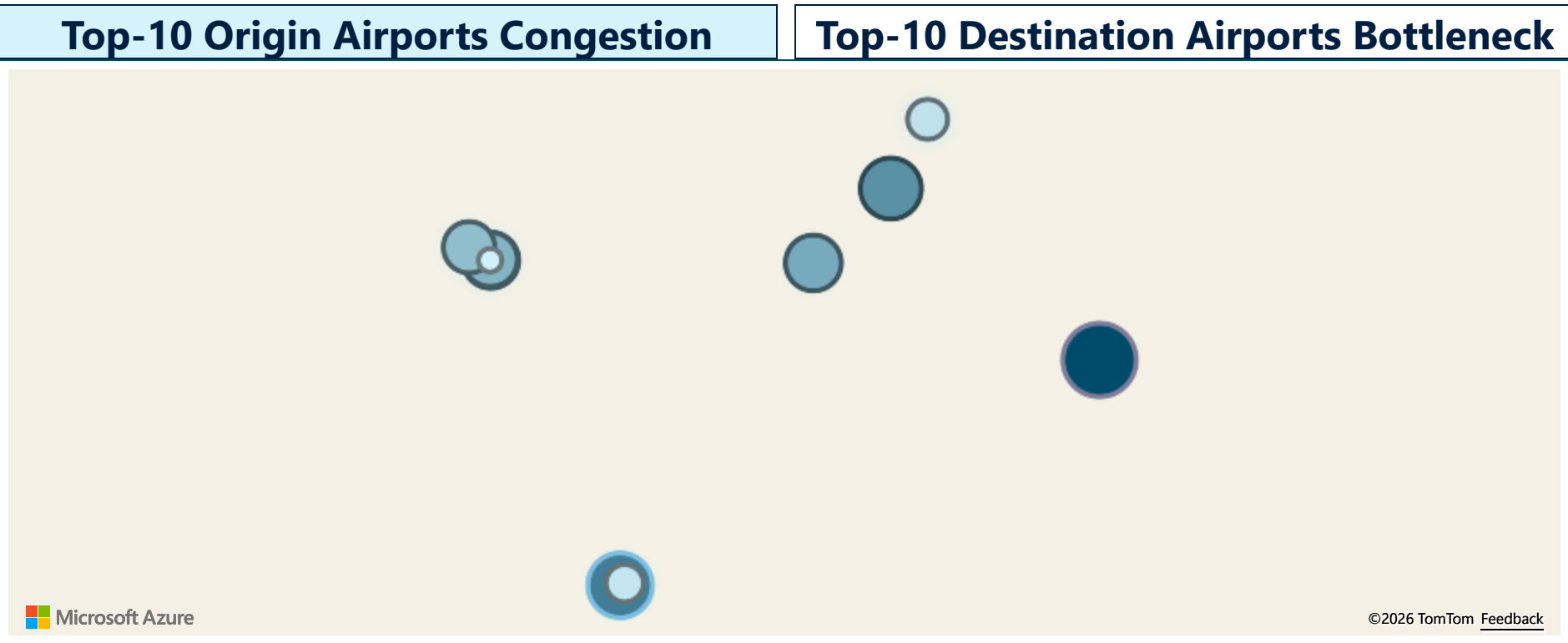
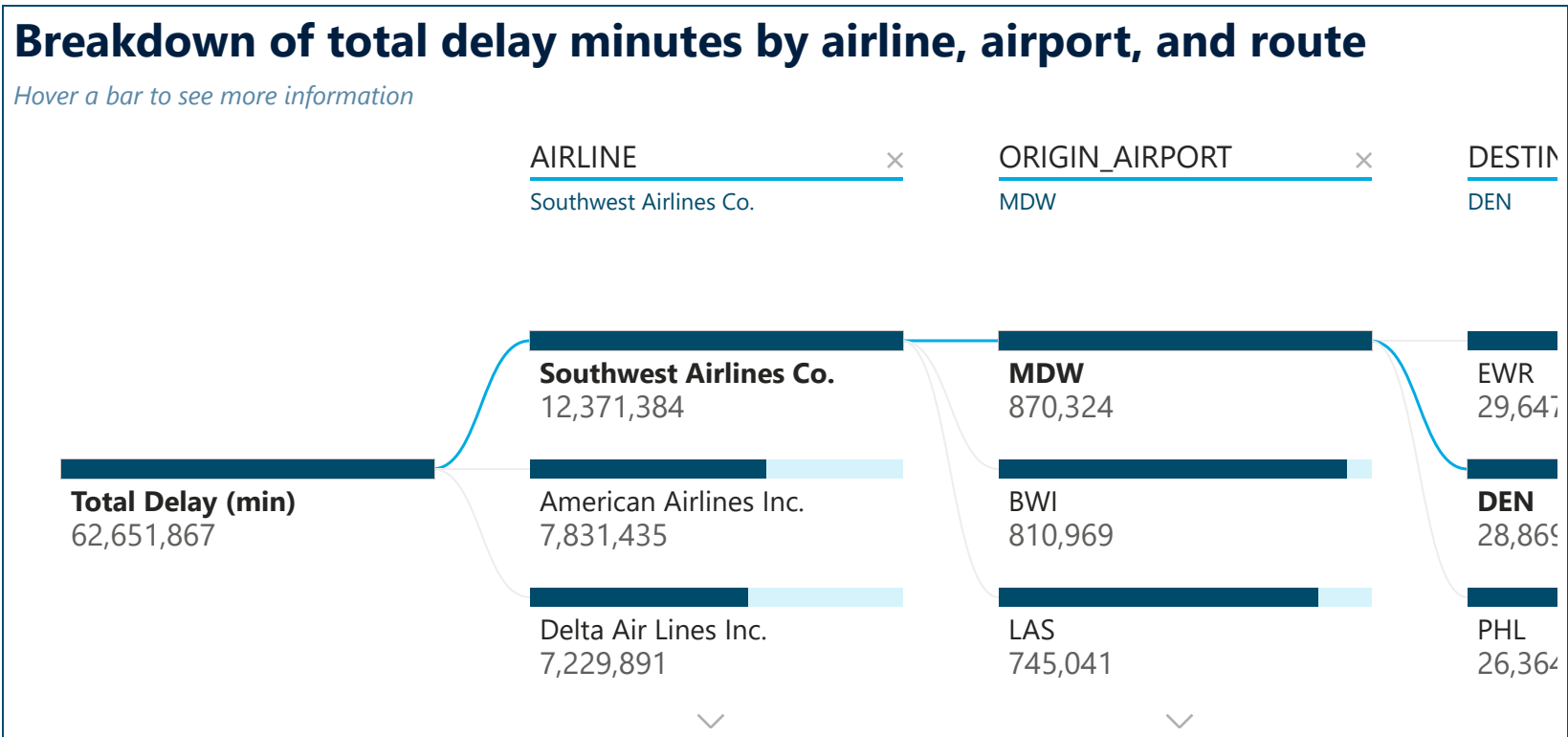
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Which airlines perform well and which requires operational review?

All airlines -> All airports (Drill down to see more)

AIRLINE	On-time Rate	Delayed Flights	Avg Arrival Delay (min)	Avg Departure Delay (min)	Delay Rate	Cancellation Rate
☐ American Eagle Airlines Inc.	79.03%	58,623	64.54	56.16	20.97%	5.10%
☐ Atlantic Southeast Airlines	81.08%	105,331	64.49	58.31	18.92%	2.66%
☐ US Airways Inc.	82.05%	34,930	51.93	42.12	17.95%	2.05%
☐ Spirit Air Lines	71.25%	33,166	64.74	58.89	28.75%	1.71%
☐ Skywest Airlines Inc.	82.05%	103,850	62.08	57.07	17.95%	1.69%
☐ JetBlue Airways	78.21%	57,257	62.55	57.59	21.79%	1.60%
☐ American Airlines Inc.	82.49%	125,238	61.93	56.60	17.51%	1.50%
☐ United Air Lines Inc.	80.10%	101,303	66.38	66.81	19.90%	1.27%
☐ Southwest Airlines Co.	81.75%	227,317	53.81	53.58	18.25%	1.27%
☐ Virgin America	81.51%	11,348	57.87	52.02	18.49%	0.86%
☐ Frontier Airlines Inc.	74.68%	22,851	68.84	61.28	25.32%	0.65%





2015 Airline Delay Analysis - Detailed Delay Time

MONTH

All

AIRLINE

All

IATA_CODE

All

Delayed Flights

LM: 67.58K

▲ MoM 38.1%

1.02M

Peak Hour Delay Share

LM: 31.55%

▼ MoM 8.7%

32%

Peak hours = 16:00 – 19:59

Avg. Arrival Delay

LM: 59.13

▲ MoM 8.9%

61

(min)

Avg. Departure Delay

LM: 55.12

▲ MoM 11.1%

57

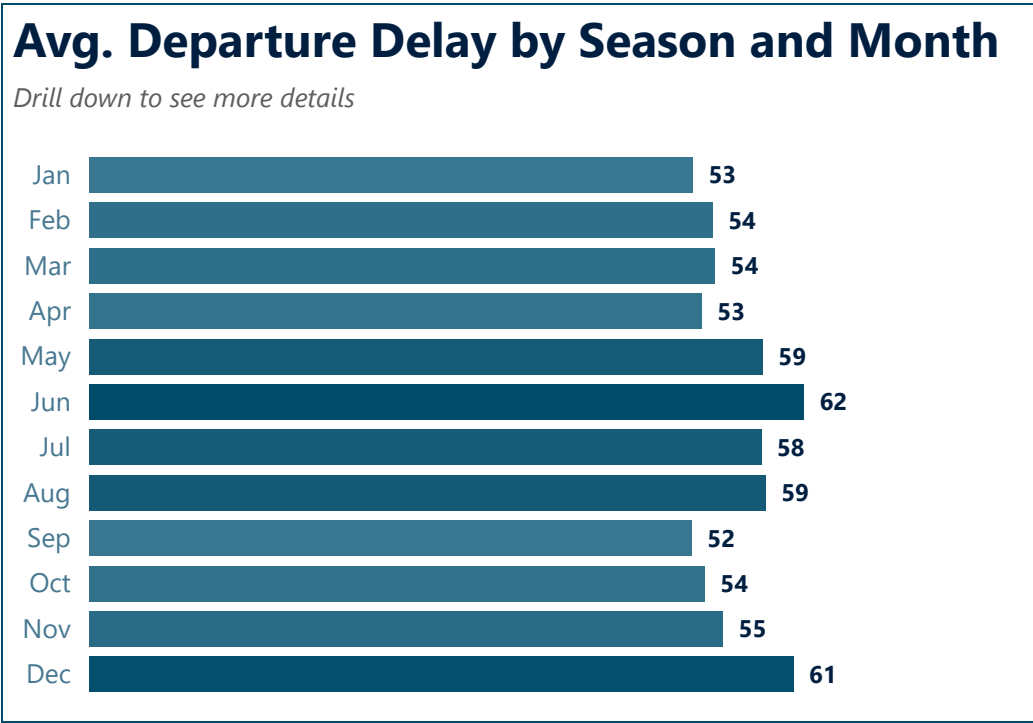
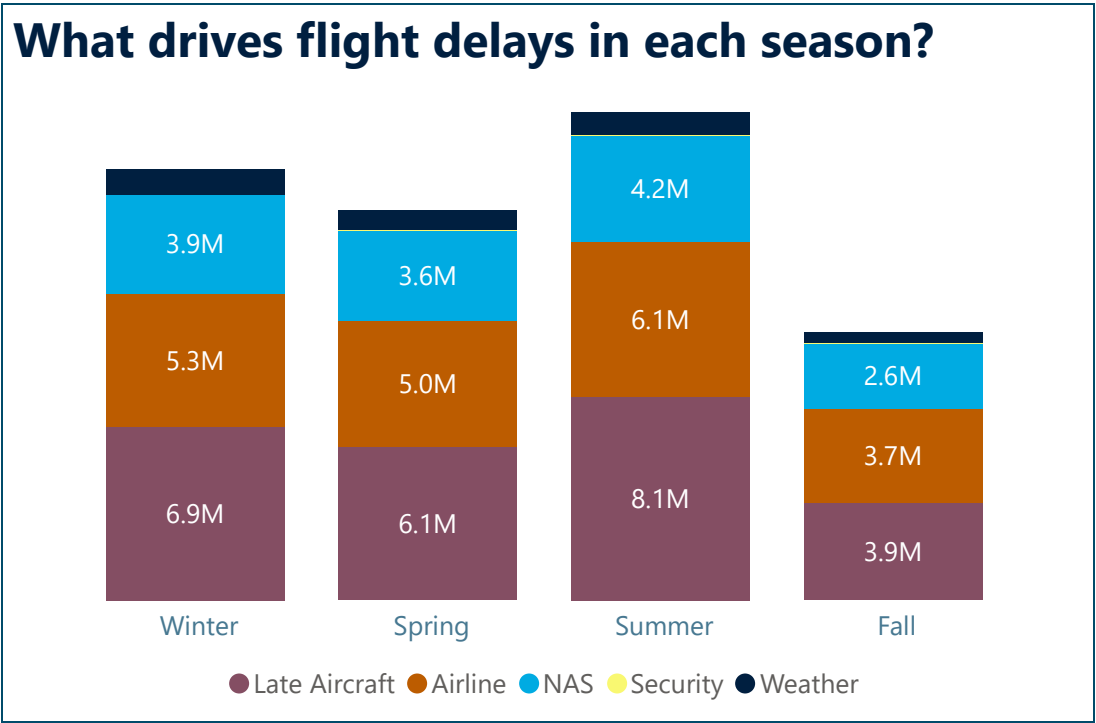
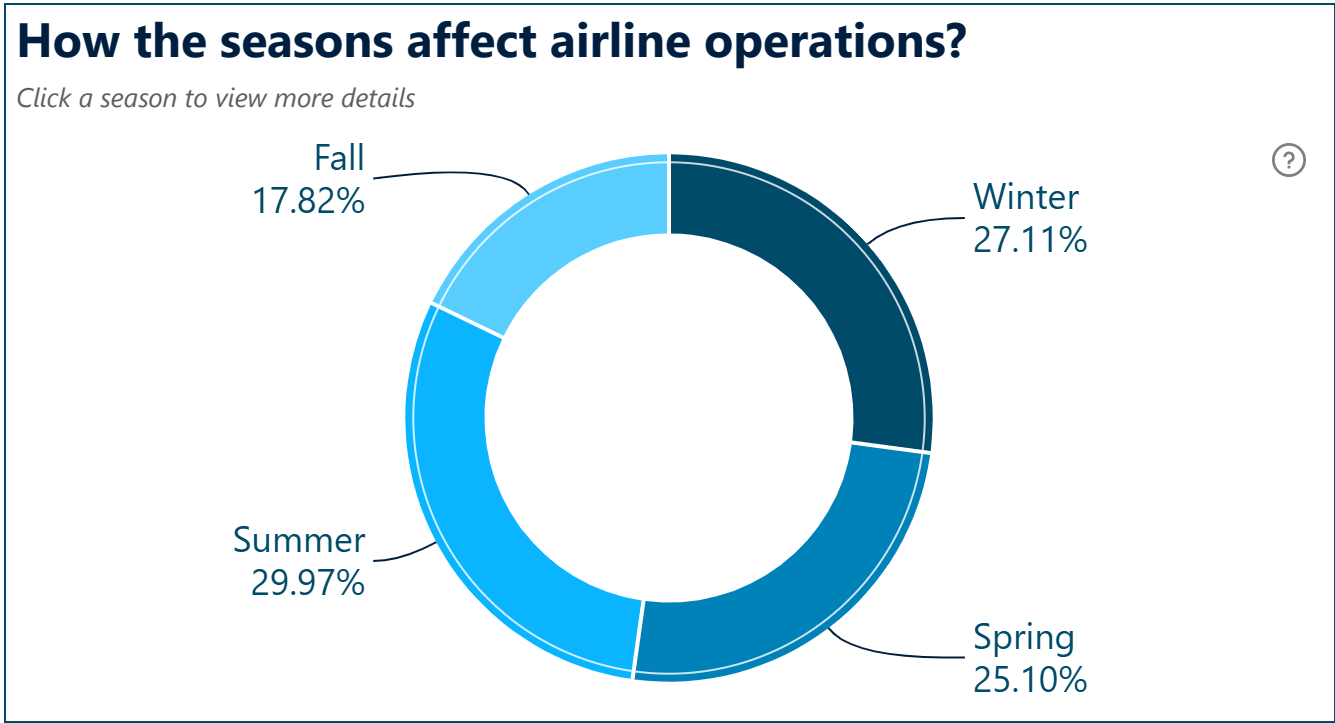
(min)

Total Delay (min)

LM: 4.04M

▲ MoM 50.0%

63M



Delay Heatmap by day of week and hour

Apply by which dimension?

Avg. Departure Delay

Avg. Arrival Delay

	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Monday	51.8	74.0	55.5	90.1	56.2	71.8	61.0	55.1	58.0	56.7	59.2	59.4	60.3	59.7	59.7	61.9	62.4	62.4	62.5	61.7	62.6	59.2	59.2	59.2
Tuesday	55.6	49.5	43.4	69.1	46.5	70.1	56.4	52.5	55.6	52.1	53.7	55.9	58.3	54.9	56.7	58.3	59.3	60.3	61.7	60.8	61.5	60.8	60.8	60.8
Wednesday	53.9	67.4	46.3	73.6	48.1	63.6	55.3	49.1	49.6	51.1	53.2	55.7	56.0	55.3	54.8	55.7	55.5	56.0	57.1	56.9	56.7	56.7	56.7	56.7
Thursday	48.8	48.5	66.9	65.7	74.7	59.3	53.0	48.3	50.9	48.9	52.1	50.7	53.9	54.3	53.9	55.8	56.2	56.6	58.5	57.5	58.2	56.2	56.2	56.2
Friday	51.8	48.3	54.8	56.5	38.3	62.0	51.9	48.1	49.1	47.9	50.3	50.5	52.5	51.9	53.5	54.6	56.0	56.5	58.5	57.0	58.8	59.2	59.2	59.2
Saturday	51.3	38.1	91.1	59.4	108.9	70.0	55.3	52.4	51.5	49.4	52.9	54.6	57.8	58.3	57.2	59.3	57.5	60.6	61.0	61.2	61.8	60.8	60.8	60.8
Sunday	50.7	59.6	73.0	72.7	58.3	62.5	62.0	58.3	54.5	54.7	56.5	55.5	57.1	56.0	56.7	58.1	57.9	58.3	59.3	57.3	59.0	57.3	57.3	57.3

Heatmap - V1.0.0.6

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